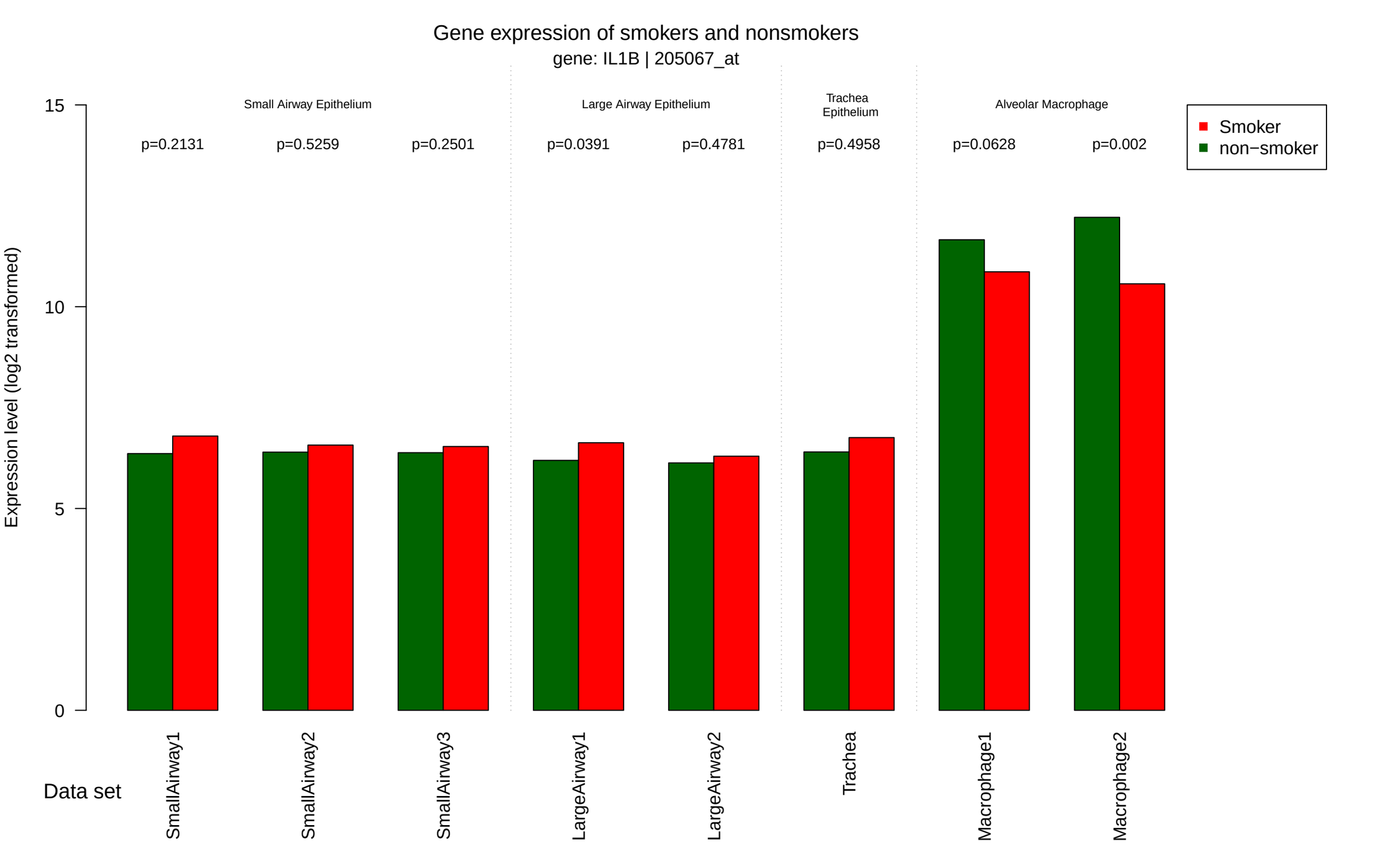
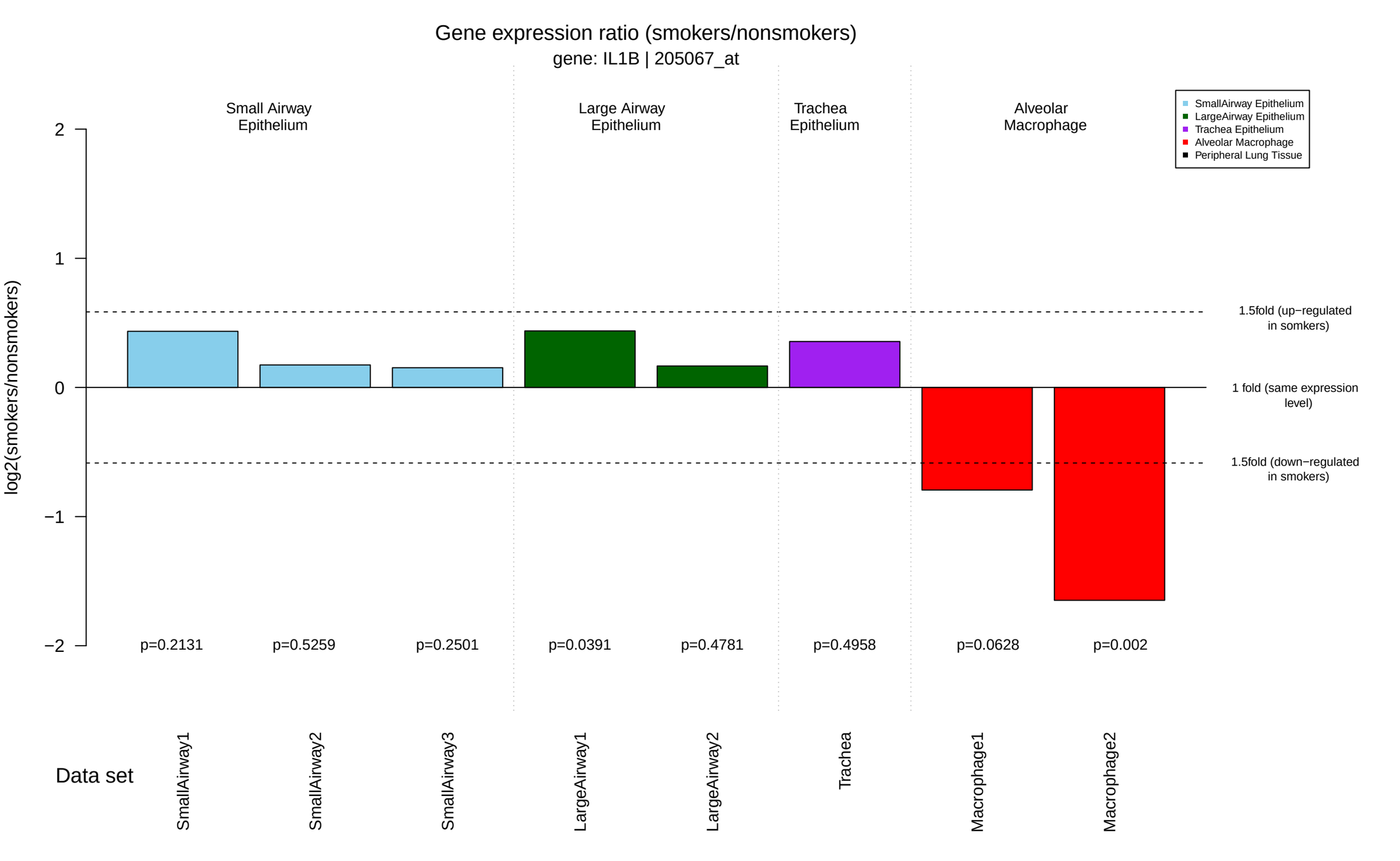
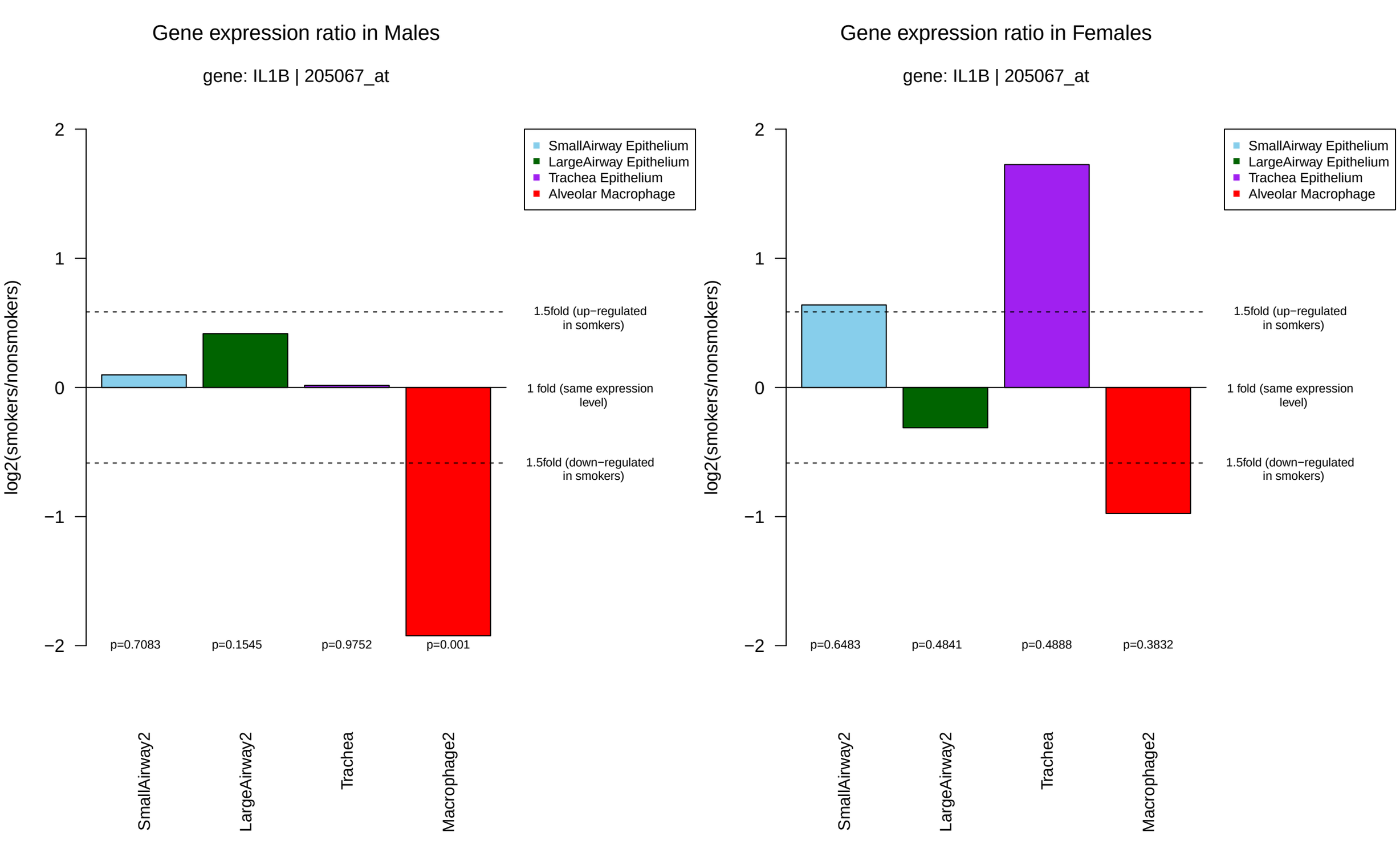
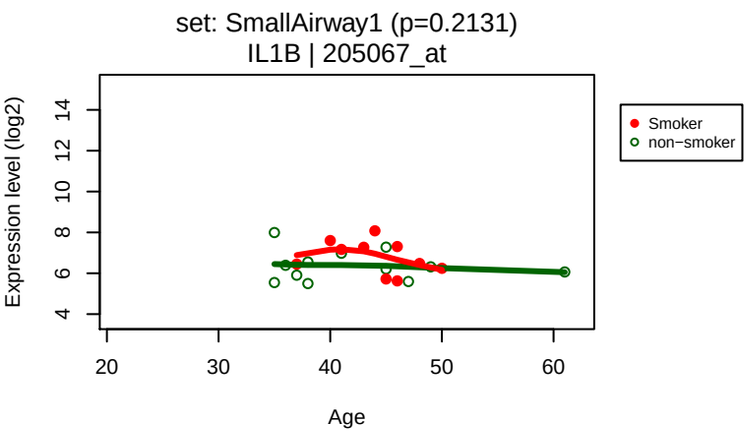
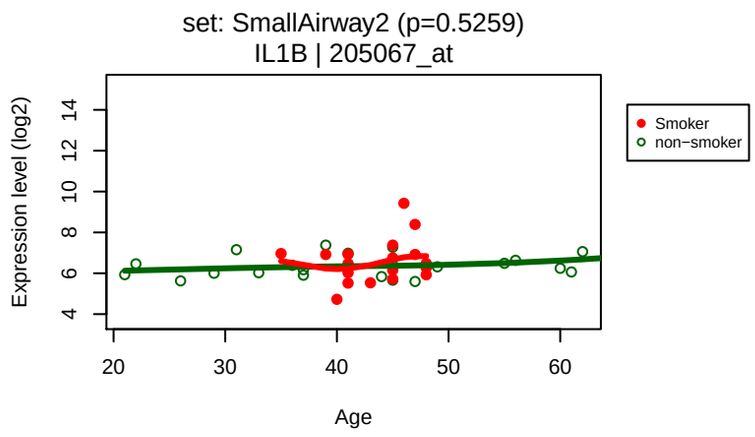
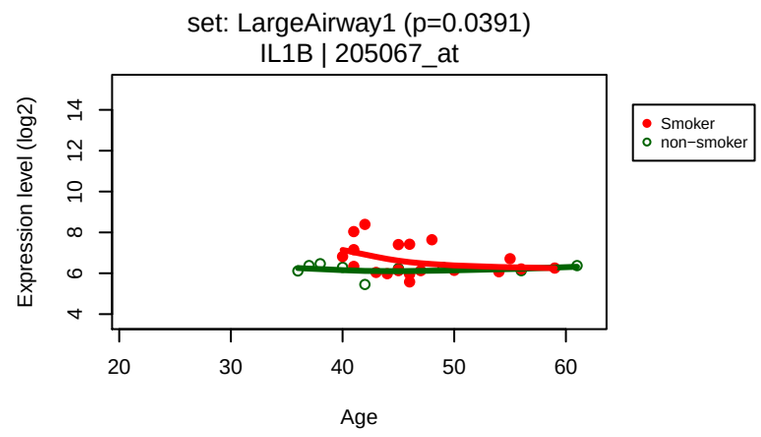

Gene expression of smokers and nonsmokers
gene: IL1B | 205067_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2131
p=0.5259
p=0.2501
p=0.0391
p=0.4781
p=0.4958
p=0.0628
p=0.002
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

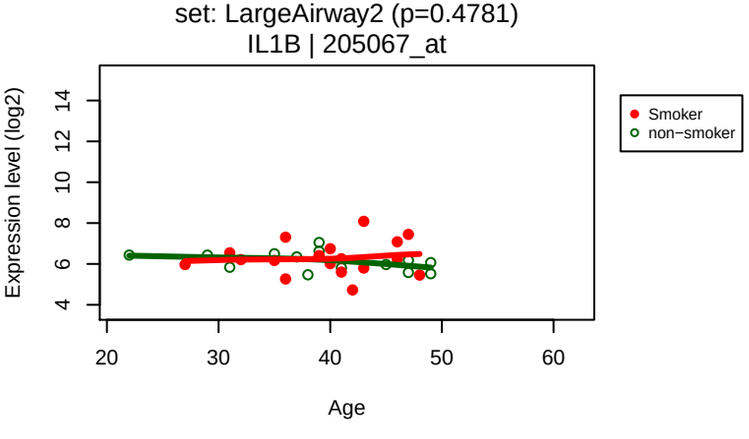

Gene expression ratio (smokers/nonsmokers)
gene: IL1B | 205067_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2131
p=0.5259
p=0.2501
p=0.0391
p=0.4781
p=0.4958
p=0.0628
p=0.002
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

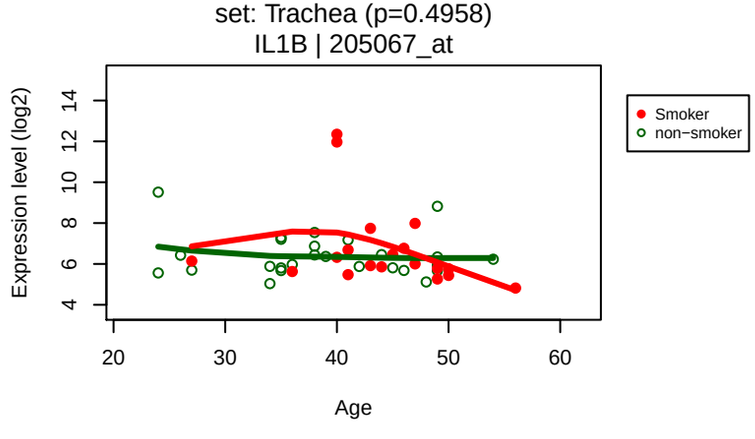

Gene expression ratio in Males
gene: IL1B | 205067_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7083
p=0.1545
p=0.9752
p=0.001
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL1B | 205067_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6483
p=0.4841
p=0.4888
p=0.3832
SmallAirway2
LargeAirway2
Trachea
Macrophage2

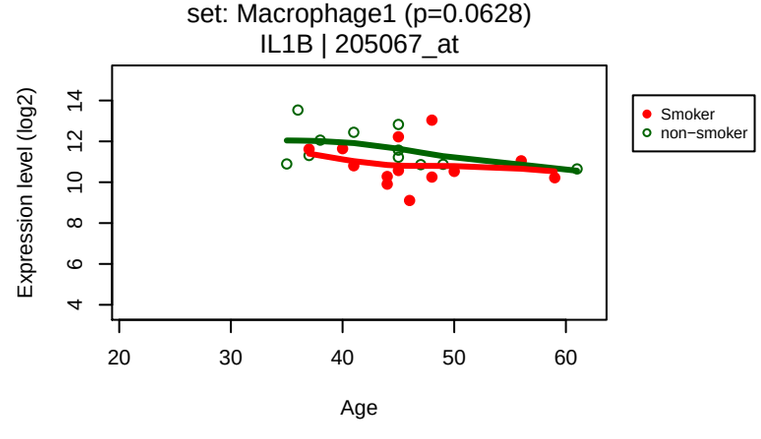

set: SmallAirway1 (p=0.2131)
IL1B | 205067_at
Expression level (log2)
Age
Smoker
non−smoker

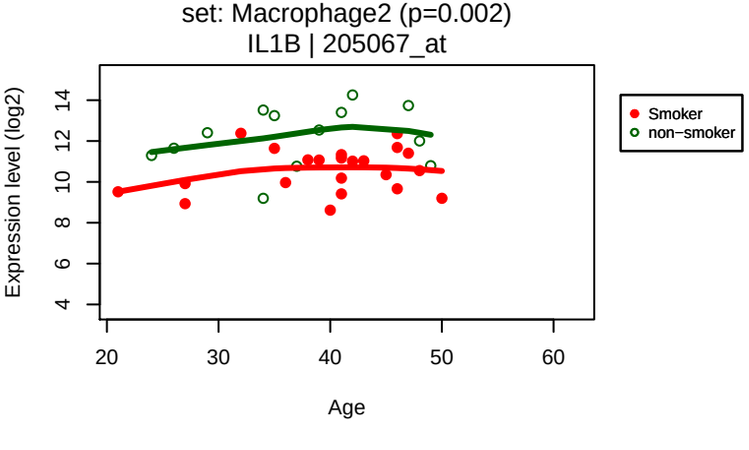

set: SmallAirway2 (p=0.5259)
IL1B | 205067_at
Expression level (log2)
Age
Smoker
non−smoker

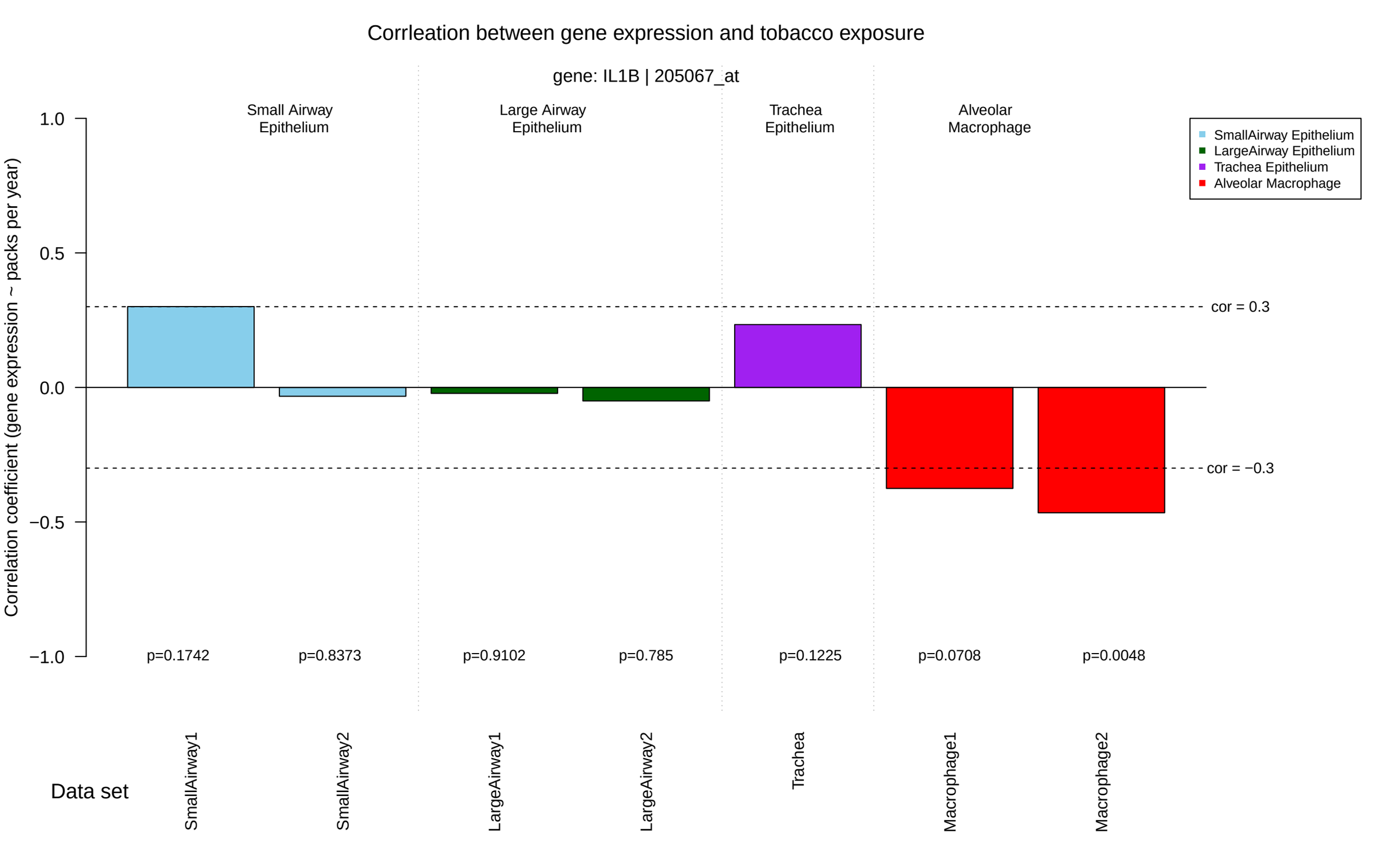

set: LargeAirway1 (p=0.0391)
IL1B | 205067_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4781)
IL1B | 205067_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4958)
IL1B | 205067_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0628)
IL1B | 205067_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.002)
IL1B | 205067_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL1B | 205067_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1742
p=0.8373
p=0.9102
p=0.785
p=0.1225
p=0.0708
p=0.0048
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2